

UPPSALA, Sweden

WMO: 024580

Lat: 59.897N

Lon: 17.589E

Elev: 21

StdP: 101.08

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.0	-15.6	-21.1	0.6	-18.7	-17.6	0.8	-15.3	10.8	1.8	9.6	0.6	0.9	310	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	27.2	18.8	25.2	17.7	23.4	16.8	19.9	24.8	18.7	23.5	17.7	21.9	3.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	21.8	17.0	12.1	20.5	16.0	11.4	19.6	56.8	24.8	53.2	23.7	50.0	21.9	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.3	7.4	-22.8	29.4	4.6	2.0	-26.1	30.8	-28.8	32.0	-31.4	33.1	-34.7	34.5		
(5)			-22.9	21.0	4.5	1.7	-26.1	22.2	-28.8	23.2	-31.3	24.1	-34.6	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-3.2	-3.5	-0.1	5.1	10.0	14.2	17.0	15.9	11.6	6.3	1.9	-1.4	
	DBStd	8.26	5.48	5.53	4.64	3.59	3.71	3.00	2.80	2.85	3.18	3.67	4.06	5.35	
	HDD10.0	2042	409	377	313	152	47	3	0	1	20	124	243	353	
	HDD18.3	4457	667	610	571	397	259	128	60	84	202	373	493	612	
	CDD10.0	660	0	0	0	5	46	129	218	184	68	9	0	0	
	CDD18.3	35	0	0	0	0	1	4	20	10	1	0	0	0	
	CDH23.3	402	0	0	0	0	11	54	238	98	2	0	0	0	
	CDH26.7	70	0	0	0	0	1	4	54	11	0	0	0	0	
Wind	WSAvg	3.5	3.6	3.6	3.6	3.7	3.6	3.5	3.0	3.1	3.4	3.5	3.5	3.7	
Precipitation	PrecAvg	570	40	28	29	33	36	49	72	70	58	53	55	46	
	PrecMax	764	79	74	55	94	105	118	147	183	144	125	125	112	
	PrecMin	401	4	3	0	3	3	11	6	15	6	11	10	12	
	PrecStd	92	18	16	14	19	23	26	35	41	32	29	28	22	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	7.6	13.3	19.9	24.7	27.2	30.5	28.0	22.8	16.1	11.6	9.9	
		MCWB	5.7	5.7	8.3	12.1	15.9	18.3	19.9	18.7	16.3	13.1	10.2	8.5	
	2%	DB	5.5	5.7	10.1	16.9	22.5	24.9	28.2	26.1	20.6	14.2	9.9	7.7	
		MCWB	4.6	4.1	6.0	10.4	14.5	16.9	19.3	18.6	15.7	12.3	9.0	6.7	
	5%	DB	4.3	4.5	7.7	14.4	19.9	22.9	26.1	24.0	18.8	13.0	8.7	6.2	
		MCWB	3.5	3.2	4.9	9.2	13.1	15.9	18.4	17.7	14.8	11.5	7.9	5.4	
	10%	DB	3.3	3.3	5.8	12.1	17.5	20.9	24.0	22.1	17.1	11.8	7.5	4.8	
		MCWB	2.5	2.1	3.5	7.5	11.7	14.8	17.4	16.8	13.9	10.5	6.8	4.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	6.0	8.3	12.9	17.0	19.3	21.7	21.3	17.8	13.9	10.5	8.6
			MCDB	6.8	7.0	12.5	18.6	23.3	25.1	27.5	25.6	21.2	15.2	11.1	9.5
2%		WB	4.7	4.4	6.5	11.0	15.3	17.7	20.3	19.6	16.3	12.7	9.1	6.8	
		MCDB	5.2	5.4	9.5	15.8	20.6	23.2	25.8	23.7	19.3	13.8	9.7	7.4	
5%		WB	3.7	3.3	5.1	9.5	13.7	16.6	19.2	18.4	15.3	11.8	7.9	5.4	
		MCDB	4.2	4.3	7.2	14.0	19.2	21.7	24.5	22.3	18.1	12.9	8.5	6.0	
10%		WB	2.6	2.3	3.9	8.0	12.3	15.5	18.1	17.5	14.3	10.6	6.9	4.2	
		MCDB	3.2	3.1	5.5	11.4	16.5	19.9	22.8	21.0	16.6	11.6	7.4	4.7	
Mean Daily Temperature Range	5% DB	MDBR	5.0	6.2	7.4	9.7	10.5	9.8	10.4	10.0	8.6	6.2	4.4	4.7	
		MCDBR	4.6	5.4	9.8	13.9	15.3	13.8	14.7	13.3	10.8	7.1	5.4	4.7	
		MCWBR	4.3	4.5	6.5	8.4	8.4	7.1	6.9	6.6	6.5	5.5	5.0	4.4	
	5% WB	MCDBR	4.6	4.9	9.0	12.9	13.4	11.9	12.4	10.9	9.5	6.3	5.2	4.5	
		MCWBR	4.4	4.2	6.3	8.0	7.8	6.6	6.3	6.1	6.0	5.1	4.9	4.3	
Clear-Sky Solar Irradiance	taub	0.273	0.315	0.336	0.371	0.369	0.360	0.385	0.377	0.343	0.331	0.299	0.256		
	taud	2.198	2.273	2.351	2.327	2.391	2.443	2.392	2.412	2.497	2.422	2.250	2.121		
	Ebn at Noon	506	683	795	827	859	873	837	815	788	667	466	368		
	Edh at Noon	42	68	86	105	106	102	105	96	75	59	40	30		
All-Sky Solar Radiation	RadAvg	0.29	0.93	2.43	3.91	5.06	5.64	5.20	4.10	2.67	1.20	0.38	0.19		
	RadStd	0.04	0.14	0.30	0.38	0.48	0.37	0.52	0.41	0.31	0.20	0.09	0.02		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	-1.02	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-113	-145	N/A	N/A

Nomenclature: See separate page